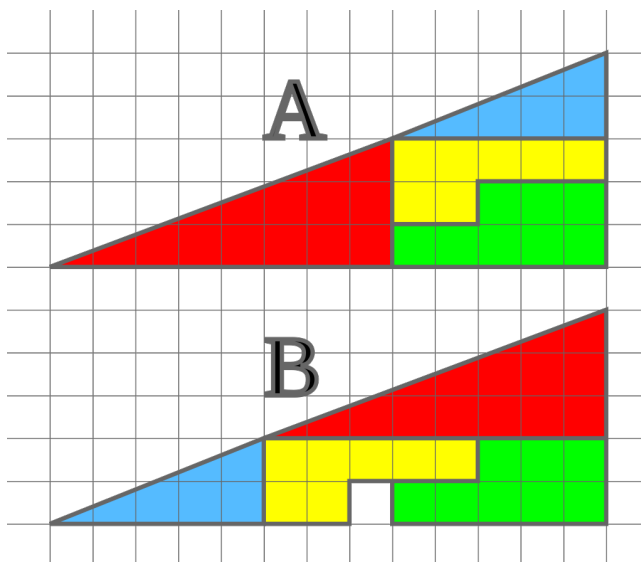


Twelfth day of class - 10/2

Fun fact: You can take a right triangle, break it into four pieces, and rearrange the pieces to decrease the area:



(Image credit: Wikipedia)

Last time: Magma demo with linear algebra, for loops, and computing the group of points on an elliptic curve (over $\mathbb{Q}$ and over a finite field).

Section 2.2 - Real and complex points on cubic curves

Facts:

- Suppose that G is an abelian group and H is a subgroup of G . There is a quotient group G/H whose elements are the set of cosets of H in G .
- The map $\phi : G \rightarrow G/H$ given by $\phi(a) = a + H$ is a surjective homomorphism. The kernel of ϕ is H .
- This construction produces *all* surjective homomorphisms!

Theorem (First isomorphism theorem). *Suppose that G is any abelian group and $\phi : G \rightarrow K$ is a surjective homomorphism. Let $H = \ker \phi$. Then $G/H \cong K$.*

Comment: Quotient groups are the reason for the notation $\mathbb{Z}/n\mathbb{Z}$ for the cyclic group of order n .

Back to elliptic curves

We have the homomorphism $\phi : \mathbb{C} \rightarrow E(\mathbb{C})$. The kernel of ϕ is L , and so $E(\mathbb{C}) \cong \mathbb{C}/L$.

The elements of $\mathbb{C}/L$ are cosets $z + L$. Another way of thinking about this is that we can think of

$$\mathbb{C}/L = \{a\omega_1 + b\omega_2 : 0 \leq a < 1, 0 \leq b < 1\}.$$

Then for $u, v \in \mathbb{C}/L$, we define $u + v$ to be the result of adding u and v , and then subtracting $0, \omega_1, \omega_2$, or $\omega_1 + \omega_2$ to get back into the set above. The set $\{a\omega_1 + b\omega_2 : 0 \leq a < 1, 0 \leq b < 1\}$ is called the *fundamental parallelogram*.

Draw a picture.

Q: If $P_1 = (0 : 1 : 0)$, P_2 , P_3 and P_4 are the points of order dividing 2 in $E(\mathbb{C})$, what do they correspond to under the map ϕ ?

A: Well, $0 \in L$ and so $\phi(0) = (0 : 1 : 0) = P_1$. Since the set L is the identity element in $\mathbb{C}/L$, we have that $z \in \mathbb{C}/L$ has order 2 if and only if $2z \in L$ but $z \notin L$. This means that $2z = a\omega_1 + b\omega_2$ and at least one of a or b is odd, so

$$z = \frac{a}{2}\omega_1 + \frac{b}{2}\omega_2.$$

Now, when we “reduce” mod L , we can assume that $0 \leq a/2 < 1$ and $0 \leq b/2 < 1$. This means that $(a/2, b/2) = (1/2, 0), (0, 1/2)$ or $(1/2, 1/2)$. So the points of order 2 in $\mathbb{C}/L$ are $\frac{\omega_1}{2}, \omega_2/2$ and $\frac{\omega_1 + \omega_2}{2}$.

Corollary. *If $m \geq 1$, then there are precisely m^2 points in $\mathbb{C}/L$ with order dividing m . Hence, $E[m](\mathbb{C}) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$.*

In fact, if P is any point in $E(\mathbb{C})$, then there are precisely m^2 points Q so that $mQ = P$.

Example 1: Let $E : y^2 = x^3 - 13392x - 1080432$.

Magma can determine the lattice corresponding to E . (The command is `Periods(E)`. It gives you two numbers: ω_1 and ω_2 .)

We find that $\omega_1 \approx 0.211535$ and $\omega_2 \approx 0.105767 + 0.243136i$.

The point $P = (168, 1188)$ is a point of order 5 in $E(\mathbb{Q})$, and in fact $E(\mathbb{Q}) = \langle P \rangle$.

We have that $\phi(z) = P$, where $z \approx -0.08461395$.

(The Magma command is `EllipticLogarithm(E! [168, 1188])`. This always returns a value of $z = x + iy$ where $-\omega_1/2 \leq x < \omega_1/2$ and $-\text{Im}(\omega_2)/2 \leq y < \text{Im}(\omega_2)/2$.)

We find that $z \equiv \frac{3}{5}\omega_1$. Now, $2z = -0.1692279 \equiv \frac{6}{5}\omega_1 \equiv \frac{1}{5}\omega_1 \pmod{L}$.

It turns out that $\phi(\omega_2/5)$ maps to the point

$$(-247.623 - 255.831i, 6338.187 - 2121.300i).$$

(You can get this with `EllipticExponential(E, Periods(E) [2]/5)`.)

With more decimals, it is possible to see that this is the point

$$(72\zeta^3 - 72\zeta^2 - 180\zeta - 192, -1404\zeta^3 - 2160\zeta^2 - 648\zeta + 486),$$

where $\zeta = e^{2\pi i/5}$. The x and y coordinates of this point satisfies a degree 4 polynomial with integer coefficients. These polynomials can be found by using the group law on the elliptic curve.

Example 2: Let $E : y^2 = x^3 - 16x + 16$.

We have $\omega_1 \approx 1.4967293231159798149$ and $\omega_2 \approx 1.22569i$. The point $P = (4 : -4 : 1)$ is a point of infinite order on E .

We have $\phi(z) = P$, where $z \approx 0.56713660783$.

Claim: z/ω_1 is irrational.

If it were rational, then $z = (a/b)\omega_1$ for some integers a and b . But then $bz = a\omega_1 \in L$ and this would mean that

$$\phi(bz) = (0 : 1 : 0).$$

But since ϕ is a homomorphism, $\phi(bz) = b\phi(z)$ and so $P = (4 : -4 : 1) = \phi(z)$ would have order b . But this is a contradiction because P has infinite order.

If n is a positive integer, and nz/ω_1 is close to an integer k , then nz is close to $k\omega_1$. Since $\wp_L(z)$ has a pole at points in L , this means that $\wp_L(nz)$ will be very large.

For example, we find that

$$351z = 132.9999661\omega_1,$$

and this means that $351P$ has coordinates that are large. In fact,

$$351P \approx (390170700.0433 : 7706940023561.8368 : 1).$$

Section 2.3 - The discriminant

If $f(x) = x^3 + ax^2 + bx + c$, is a polynomial with integer coefficients, the *discriminant* of f is defined by

$$D = -4a^3c + a^2b^2 + 18abc - 4b^3 - 27c^2.$$

Theorem (Lutz-Nagell). *Suppose that $E : y^2 = x^3 + ax^2 + bx + c$ is an elliptic curve with $a, b, c \in \mathbb{Z}$ and $P = (x, y)$ is a point of finite order on E . Then*

- (1) $x, y \in \mathbb{Z}$, and
- (2) $y = 0$ or $y|D$.

Note: This is false for elliptic curves in long Weierstrass form.

Example:

Let $E : y^2 = x^3 - 2x^2 - x + 3$. We have $a = -2$, $b = -1$ and $c = 3$ and so

$$D = -31.$$

We use Magma to find that $x^3 - 2x^2 - x + 3$ is irreducible, so there are no rational points with $y = 0$.

We test each divisor y of 31 (which is only $y = 1$ and $y = 31$), and factor $x^3 - 2x^2 - x + 3 - y^2$. (If the discriminant was larger, we could use the Magma command `Divisors(D)`; to return a list of all the divisors of the discriminant.)

For $y = 1$ we find three roots: $x = -1$, $x = 1$ and $x = 2$. This gives the points $(2, \pm 1)$, $(1, \pm 1)$, $(-1, \pm 1)$.

For $y = 31$, there are no rational roots of $x^3 - 2x^2 - x + 3 - 31^2$. So we don't get any additional points.

If we let $P_1 = (2, 1)$, we find that $2P_1 = (1/4, 13/8)$ is not integral, so P_1 is not a torsion point.

If $P_2 = (1, 1)$, we find $2P_2 = (1, -1) = -P_2$, so P_2 is a point of order 3.

If $P_3 = (-1, 1)$, we find that $2P_3 = (13, -43)$ and $3P_3 = (-6/49, 603/343)$, so P_3 is not a torsion point.

It follows that the set of rational points of finite order on E is $\{(0 : 1 : 0), (1 : 1 : 1), (1 : -1 : 1)\}$.

(In fact, $E(\mathbb{Q}) \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}$ and is generated by $(1 : 1 : 1)$ and $(-1 : 1 : 1)$.)

Properties of the discriminant

If $f(x) = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3)$, the discriminant of f is defined by

$$D = (\alpha_1 - \alpha_2)^2(\alpha_1 - \alpha_3)^2(\alpha_2 - \alpha_3)^2.$$

(This is analogous to how the discriminant of $x^2 + bx + c$, namely $b^2 - 4c$ is defined. It is $(\alpha_1 - \alpha_2)^2$, where α_1 and α_2 are the roots of $x^2 + bx + c$.)

- $D = 0$ if and only if $f(x)$ has repeated roots. Repeated roots of $f(x)$ make $y^2 = f(x)$ singular, so $y^2 = f(x)$ is non-singular if and only if $D \neq 0$.
- The same property holds modulo p , that $f(x)$ has no repeated roots in $\mathbb{F}_p$ if and only if $p \nmid D$. So an elliptic curve $E/\mathbb{F}_p$ is singular precisely for the primes p that divide D . (There's always at least one such p !)
- If all the roots of the cubic are real (which corresponds to $E : y^2 = x^3 + ax^2 + bx + c$ having two pieces), then $D > 0$. If one root of the cubic is real, and two are complex, then $D < 0$ (which corresponds to $E : y^2 = x^3 + ax^2 + bx + c$ having one piece).

Where does the formula $D = -4a^3c + a^2b^2 + 18abc - 4b^3 - 27c^2$ come from?

We have $a = -(\alpha_1 + \alpha_2 + \alpha_3)$, $b = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3$ and $c = -\alpha_1\alpha_2\alpha_3$. These are the elementary symmetric polynomials in the roots of $f(x)$. A polynomial is *symmetric* if permuting the variables doesn't change the polynomial, and every symmetric polynomial is a polynomial in these elementary symmetric polynomials. This is why the identity is true. Verifying it is a little bit tedious.

Fact: We have that f has a repeated root if and only if $f(x)$ and $f'(x)$ share a common root. When they don't we can say that the "gcd" of $f(x)$ and $f'(x)$ is 1 (as a polynomial).

We can use the Euclidean algorithm to find polynomials with rational coefficients $a(x)$ and $b(x)$ so that $a(x)f(x) + b(x)f'(x) = 1$.

Multiplying through by the denominators in the expressions for $a(x)$ and $b(x)$ gives the identity below:

$$\begin{aligned} D &= [(18b - 6a^2)x - (4a^3 - 15ab + 27c)] f(x) \\ &\quad + [(2a^2 - 6b)x^2 + (2a^3 - 7abx + 9c)x + (a^2b + 3ac - 4b^2)] f'(x). \end{aligned}$$

Proof of Lutz-Nagell (part b)

Lemma: Let $P = (x, y) \in E(\mathbb{Q})$ be such that P and $2P$ both have integer coordinates. Then either $y = 0$ or $y|D$.

Proof: Let $2P = (X, Y)$. By the duplication formula, we have

$$2x + X = \lambda^2 - a,$$

where $\lambda = \frac{f'(x)}{2y}$. Now, $\lambda \in \mathbb{Q}$ and $x, X, a \in \mathbb{Z}$ and so $\lambda = \sqrt{2x + X + a}$ must be an integer as well. Thus, $2y$ divides $f'(x)$ and in particular, $y|f'(x)$. Since $y^2 = f(x)$, we also have that $y|f(x)$. Thus, y divides the left hand side of

$$r(s)f(x) + s(x)f'(x) = D$$

and hence $y|D$. QED Lemma

Homework problem: Show that if (x, y) is a point of finite order on E , then $y = 0$ or $y^2|D$.